Experiment 8

Student Name: Analava Bera
UID: 23BCS13611
Branch: CSE
Section/Group: KRG 3-A
Semester: 6th
Date of Performance:10/04/2026
Subject Name: System Design
Subject Code: 23CSH-314

1. Aim : To design a stock trading platform (similar to Zerodha, Groww & Upstox) that enables users to trade stocks, view real-time market data, manage portfolios, and execute buy/sell orders with high consistency and low latency..

2. Objective:

- To design a scalable architecture for a stock trading platform.
- To enable real-time stock price updates using WebSockets.
- To support order placement, modification, and cancellation.
- To ensure high consistency and correctness in financial transactions.
- To implement portfolio tracking and watchlist features.

3. Tools Used:

• Frontend (ReactJS, HTML, CSS, JavaScript, WebSocket) o Built responsive UI with real-time stock updates

- • Backend (Node.js / Java / Spring Boot) o Developed RESTAPIs for trading, authentication,and portfolio
- • Database (PostgreSQL / MySQL) oManagedfinancial transactions and user data
- • Redis o Usedforcachingstock pricesandimproving latency
- • Message Systems / WebSocketoEnabledreal-time updates forstockpricesand order status
- • Deployment (AWS / GCP, Load Balancer, Docker) o Scalable cloudinfrastructure with containerization

4. System Requirements:

A. Functional Requirements

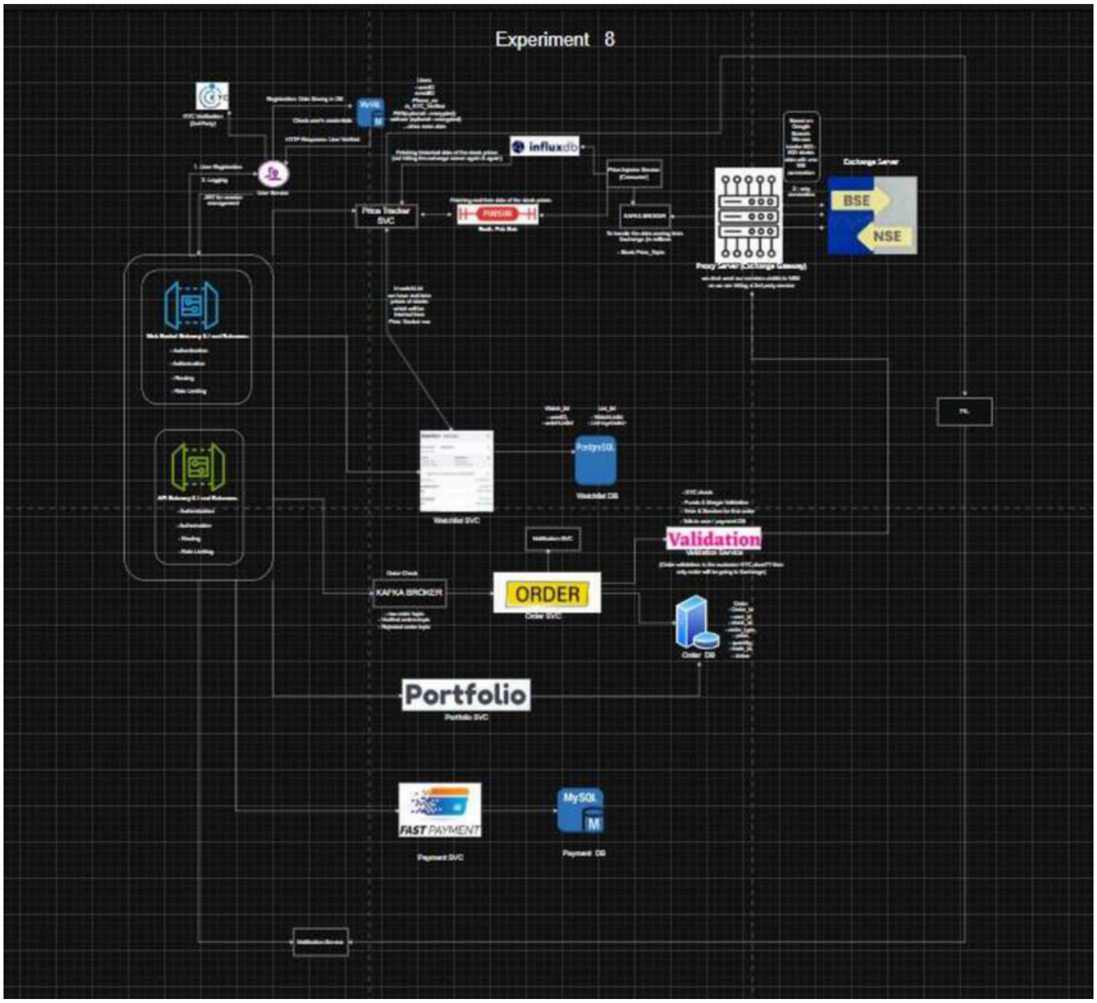
- Users should be able to register, login, and complete KYC verification.

- Users should be able to view real-time stock prices and historical data.
- Users should be able to place, modify, and cancel buy/sell orders.
- The system should support limit orders and market orders.
- Users should be able to maintain a watchlist with real-time updates.
- Users should be able to view portfolio, holdings, and performance.
- The system should provide order status updates in real-time.
- Users should be able to deposit and withdraw funds.

B. Non-Functional Requirements

- Scalability
  Systemshouldsupport 2–3 million daily active users with high-frequency trading and thousandsof stocks.
- CAP Theorem
  Consistency >>> Availability
  Financialcorrectnessiscritical(noduplicatetrades,nowrong balances).
- Latency:
  Order placement latency shouldbe < 100 ms
  Stockprice updates latencyshould be < 50 ms
- Re liabilit y:
  Systemmustensure faulttolerance and no data loss in transactions.

5. Low Level Design (LLD):



6. Scalability Solution

- Use WebSockets forreal-time stock price updates
- Implement Redis caching for frequently accessed data
- Use load balancersfor distributing traffic
- Use microservices architecture (Order, Portfolio, Market Data, Funds)
- Store transactionaldata in strongly consistent databases
- Use horizontal scaling for handling millions of users
- Optimize read-heavyoperations using caching

7.Learning Outcomes (What I Have Learnt)

- Understood architecture of real-time stock tradingsystems
- Learned importance of consistency in financial systems
- Gained knowledge of WebSocket-based real-time updates
- Understood order lifecycle (place → match → execute → settle)
- Learned how to design scalable and low-latency systems